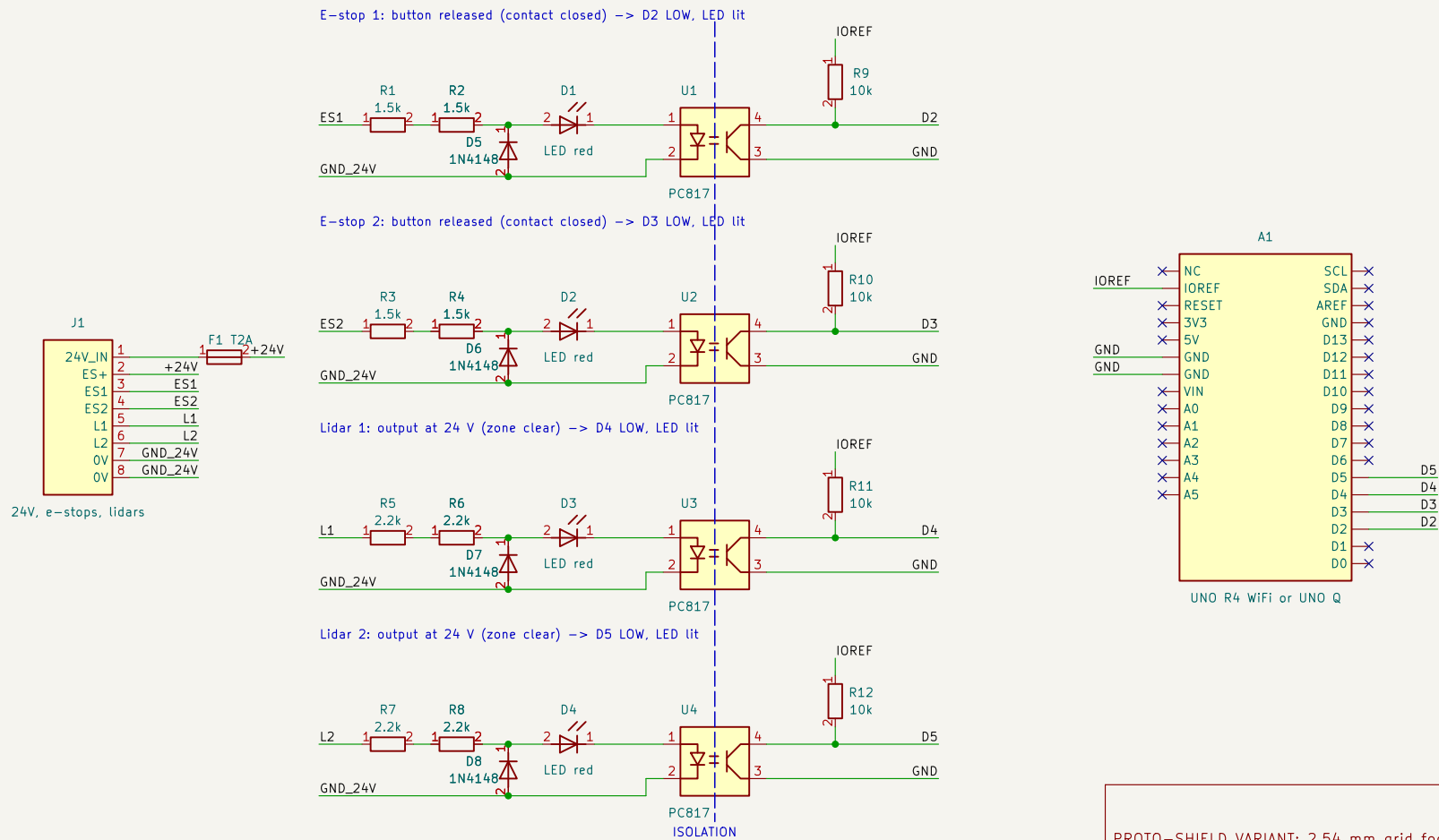


Safety inputs: 2 e-stops + 2 lidars, 24 V through PC817 – shield for Arduino UNO R4 WiFi / UNO Q

E-stop loops: $2 \times 1.5 \text{ k} \rightarrow 6.9 \text{ mA}$ (109 mW per resistor at 28.8 V). Lidar inputs: $2 \times 2.2 \text{ k} \rightarrow 4.7 \text{ mA}$ (75 mW). All resistors 1/4 W.

No current (button pressed, zone occupied, wire cut, 24 V lost) $\rightarrow$ pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons.

GND_24V and GND are NOT connected: this is the isolation barrier. Pull-ups go to IOREF (5 V on UNO R4 WiFi, 3.3 V on UNO Q), never to 5V.



PROTO-SHIELD VARIANT: 2.54 mm grid footprints, 8-way terminal
Generated by generate.py – edit the script, not this file

Sheet: /
File: pc817_bench.kicad_sch

Title: Safety inputs: 2 e-stops + 2 lidars

Size: A4 Date: 2026-10-02

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